Chapter 1. Set Theory and Logic

Sets and functions, relations, countability, recursive definition, the axiom of choice, well-ordering, Zorn's lemma

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This chapter collects the set theory used throughout. Numbers in parentheses after a result refer to the corresponding result in Munkres.

1.1 Fundamental concepts

Motivation. Topology needs very little machinery beyond sets and functions, but it uses that machinery heavily: unions and intersections of arbitrary families, images and preimages, and infinite constructions. This chapter fixes the language and proves the set-theoretic facts (countability, choice, well-ordering) that later arguments quietly rely on.

We use sets, membership, inclusion $A \subset B$, union, intersection, difference $A - B$, the empty set, and the cartesian product $A \times B$ of ordered pairs. For a collection $\mathcal{A}$ of sets, $\bigcup_{A\in\mathcal{A}} A$ and $\bigcap_{A\in\mathcal{A}} A$ are defined as usual (the intersection of the empty collection is left undefined). De Morgan's laws hold:

$$A - \bigcup_\alpha B_\alpha = \bigcap_\alpha (A - B_\alpha), \qquad A - \bigcap_\alpha B_\alpha = \bigcup_\alpha (A - B_\alpha),$$

and $A \cap \bigcup B_\alpha = \bigcup (A \cap B_\alpha)$, $A \cup \bigcap B_\alpha = \bigcap (A \cup B_\alpha)$. In logic, "$P \Rightarrow Q$" is equivalent to its contrapositive "not $Q \Rightarrow$ not P" but not to its converse; the negation of "for every x, $P(x)$" is "for some x, not $P(x)$".

1.2 Functions

A **function** $f : A \to B$ is a rule (formally a subset of $A\times B$ in which each a appears as first coordinate of exactly one pair). For $A_0 \subset A$, $f(A_0)$ is the image; for $B_0 \subset B$, $f^{-1}(B_0) = \{a : f(a) \in B_0\}$. f is **injective** if $f(a) = f(a')$ implies $a = a'$, **surjective** if $f(A) = B$, **bijective** if both; a bijection has an inverse $f^{-1} : B \to A$. Composites of injections (surjections) are injections (surjections). Preimages preserve unions, intersections and differences; images preserve unions and inclusions only, and $A_0 \subset f^{-1}(f(A_0))$, $f(f^{-1}(B_0)) \subset B_0$, with equality if f is injective (resp. surjective).

Lemma 1.1 (2.1)

Let $f : A \to B$. If there are $g, h : B \to A$ with $g(f(a)) = a$ for all $a \in A$ and $f(h(b)) = b$ for all $b \in B$, then f is bijective and $g = h = f^{-1}$.

Proof. $f(a) = f(a')$ gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; $b = f(h(b))$ shows it is surjective. Then $g = g \circ (f \circ h) = (g \circ f) \circ h = h$, and both equal f^{-1}. ∎

1.3 Relations

Motivation. Equivalence relations will produce quotient spaces, and order relations will produce the order topology on $\mathbb{R}$ and on well-ordered sets. Both are introduced here in purely set-theoretic form.

A **relation** on A is a subset $C \subset A \times A$, written xCy. An **equivalence relation** $\sim$ is reflexive, symmetric and transitive; the **equivalence class** of x is $E = \{y : y \sim x\}$.

Lemma 1.2 (3.1)

Two equivalence classes E, E' are either disjoint or equal.

Proof. If $y \in E \cap E'$ with E the class of x and E' the class of x', then $y \sim x$ and $y \sim x'$, so $x \sim x'$. Any $w \sim x$ then satisfies $w \sim x'$; so $E \subset E'$ and symmetrically. ■

Hence the classes form a **partition** of A (disjoint nonempty sets with union A); conversely every partition comes from the relation "lying in the same piece".

An **order relation** (simple or linear order) $<$ satisfies: comparability ($x \neq y \Rightarrow x < y$ or $y < x$), nonreflexivity (no $x < x$) and transitivity. In an ordered set we have open intervals (a, b), immediate successors and predecessors, and the **dictionary order** on $A \times B$: $a_1 \times b_1 < a_2 \times b_2$ iff $a_1 < a_2$, or $a_1 = a_2$ and $b_1 < b_2$. Ordered sets A, B have the same **order type** if there is an order-preserving bijection. For $A_0 \subset A$, an **upper bound** b satisfies $x \leq b$ for all $x \in A_0$; the least one is the **supremum** $\sup A_0$, and dually $\inf A_0$. A has the **least upper bound property** if every nonempty subset bounded above has a supremum; this is equivalent to the greatest lower bound property.

1.4 The integers and the real numbers

Motivation. All of analysis rests on one property of $\mathbb{R}$: every nonempty bounded-above set has a least upper bound. We record the axioms of $\mathbb{R}$ and the induction principle for $\mathbb{Z}_+$, so that later proofs (connectedness of intervals, compactness of $[a, b]$) can point to exactly what they use.

We assume $\mathbb{R}$ is an ordered field with the least upper bound property; $\mathbb{Z}_+$ is the smallest inductive subset (containing 1 and closed under $x \mapsto x + 1$), which gives the principle of induction.

Theorem 1.3 (4.1, well-ordering property)

Every nonempty subset of $\mathbb{Z}_+$ has a smallest element.

Proof. First, for each n, every nonempty subset of $\{1, \ldots, n\}$ has a smallest element: the set A of n for which this holds contains 1, and if $n \in A$ and $C \subset \{1, \ldots, n+1\}$, then either $C = \{n+1\}$ or $C \cap \{1, \ldots, n\}$ has a least element, which is the least element of C. For a nonempty $D \subset \mathbb{Z}_+$ choose $n \in D$; the least element of $D \cap \{1, \ldots, n\}$ is the least element of D. ■

Theorem 1.4 (4.2, strong induction)

Let $A \subset \mathbb{Z}_+$ be such that $\{1, \ldots, n-1\} \subset A$ implies $n \in A$, for each n. Then $A = \mathbb{Z}_+$.

Proof. Otherwise let n be the least element of $\mathbb{Z}_+ - A$; then $\{1, \ldots, n-1\} \subset A$, so $n \in A$, a contradiction. ■

The **Archimedean ordering property** ($\mathbb{Z}_+$ is unbounded above in $\mathbb{R}$) follows from the least upper bound property: if $b = \sup \mathbb{Z}_+$ existed, $b - 1 < n$ for some n, so $n + 1 > b$.

1.5 Cartesian products

For an indexed family $\{A_\alpha\}_{\alpha \in J}$, the **cartesian product** $\prod_{\alpha \in J} A_\alpha$ is the set of all functions $x : J \to \bigcup A_\alpha$ with $x(\alpha) \in A_\alpha$ (written $(x_\alpha)_{\alpha \in J}$). For finite $J = \{1, \ldots, m\}$ these are m-tuples; for $J = \mathbb{Z}_+$ they are sequences; X^J denotes the set of all functions $J \to X$, e.g. $\mathbb{R}^\omega$, the set of real sequences.

1.6 Finite sets

Motivation. "Finite" must be defined through bijections with $\{1,\dots,n\}$, and the intuitive facts (a finite set has a unique size, subsets of finite sets are finite) need proofs. The pigeonhole principle proved here is the prototype of every counting argument in the course.

A set is **finite** if it is empty or in bijection with a section $\{1,\dots,n\}$; we show n is unique.

Lemma 1.5 (6.1)

Let $a_0 \in A$. There is a bijection $A \to \{1,\dots,n+1\}$ iff there is a bijection $A - \{a_0\} \to \{1,\dots,n\}$.

Proof. Given $g : A - \{a_0\} \to \{1,\dots,n\}$, extend by $a_0 \mapsto n+1$. Conversely, given $f : A \to \{1,\dots,n+1\}$, if $f(a_0) = n+1$ restrict; otherwise compose f with the transposition of $f(a_0)$ and $n+1$ first. ∎

Theorem 1.6 (6.2)

Let $f : A \to \{1,\dots,n\}$ be a bijection and B a proper subset of A. Then there is no bijection $B \to \{1,\dots,n\}$, but (if $B \neq \emptyset$) there is a bijection $B \to \{1,\dots,m\}$ for some $m < n$.

Proof. Induction on n. For $n = 1$, A is a point and $B = \emptyset$. Assume the result for n and let $f : A \to \{1,\dots,n+1\}$, $B \subsetneq A$ nonempty. Pick $a_0 \in B$ (or, if convenient, any point); $A - \{a_0\}$ is in bijection with $\{1,\dots,n\}$ by Lemma 1.5. If $B - \{a_0\}$ is a proper subset of $A - \{a_0\}$ (true since $B \neq A$ and $a_0 \in B$), the induction hypothesis says $B - \{a_0\}$ is empty or in bijection with $\{1,\dots,p\}$, $p < n$, and not with $\{1,\dots,n\}$; by Lemma 1.5 again, B is in bijection with $\{1,\dots,p+1\}$, $p+1 < n+1$, and not with $\{1,\dots,n+1\}$. ∎

Corollary 1.7 (6.3–6.5)

A finite set has no bijection with a proper subset of itself; $\mathbb{Z}_+$ is not finite; the cardinality of a finite set is uniquely determined.

Proof. The first is Theorem 1.6. $n \mapsto n+1$ is a bijection of $\mathbb{Z}_+$ onto a proper subset. If A were in bijection with $\{1,\dots,m\}$ and $\{1,\dots,n\}$, $m < n$, then $\{1,\dots,n\}$ would be in bijection with its proper subset $\{1,\dots,m\}$. ∎

Corollary 1.8 (6.6)

A subset B of a finite set A is finite; if B is proper, its cardinality is smaller.

Corollary 1.9 (6.7)

For nonempty B the following are equivalent:
(1) B is finite;
(2) there is a surjection from some section $\{1,\dots,n\}$ onto B;
(3) there is an injection of B into some section.

Proof. (1)⇒(2) clear. (2)⇒(3): send b to the least element of $f^{-1}(b)$ (Theorem 1.3). (3)⇒(1): B is in bijection with a subset of a finite set; use Corollary 1.8. ∎

Corollary 1.10 (6.8)

Finite unions and finite cartesian products of finite sets are finite.

Proof. For $A \cup B$ with $A \cong \{1, \ldots, m\}$ and $B \cong \{1, \ldots, n\}$, the map onto $\{1, \ldots, m+n\}$ sending B to the last n values gives a surjection from a section; apply Corollary 1.9. $A \times B$ is the union of the finite sets $A \times \{b\}$; induct for more factors. ■

1.7 Countable and uncountable sets

Motivation. Many topological properties (second countability, separability) are statements about countable families. We need efficient criteria for countability, such as countable unions of countable sets, and Cantor's diagonal argument shows that not every infinite set is countable.

A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, **countable** if finite or countably infinite, and **uncountable** otherwise.

Theorem 1.11 (7.1)

For nonempty B the following are equivalent:
(1) B is countable;
(2) there is a surjection $f : \mathbb{Z}_+ \to B$;
(3) there is an injection $g : B \to \mathbb{Z}_+$.

Proof. (1)⇒(2): if B is finite, extend a surjection from a section by a constant. (2)⇒(3): $g(b) = \min f^{-1}(b)$. (3)⇒(1): $g(B) \subset \mathbb{Z}_+$ is countable by the next lemma. ■

Lemma 1.12 (7.2)

An infinite subset $C \subset \mathbb{Z}_+$ is countably infinite.

Proof. Define $h(1) = \min C$ and $h(n) = \min\{C - h(\{1, \ldots, n-1\})\}$, which is nonempty because C is infinite (recursive definition, Theorem 1.21). h is injective (increasing). It is surjective: if $c \in C$ were not in the image, then $h(n) < c$ for all n (since $h(n)$ is the least element not yet chosen and c is available), so h would map $\mathbb{Z}_+$ injectively into $\{1, \ldots, c\}$, contradicting Corollary 1.7. ■

Corollary 1.13 (7.3)

A subset of a countable set is countable.

Corollary 1.14 (7.4)

$\mathbb{Z}_+ \times \mathbb{Z}_+$ is countably infinite.

Proof. $(n, m) \mapsto 2^n 3^m$ is injective by unique factorization; apply Theorem 1.11. ■

Theorem 1.15 (7.5)

A countable union of countable sets is countable.

Proof. Let $\{A_n\}_{n\in J}$, $J \subset \mathbb{Z}_+$, be countable and nonempty. Choose surjections $f_n : \mathbb{Z}_+ \to A_n$ (and, if J is finite, a surjection $g : \mathbb{Z}_+ \to J$). Then $(k, m) \mapsto f_{g(k)}(m)$ maps $\mathbb{Z}_+ \times \mathbb{Z}_+$ onto $\bigcup A_n$; compose with a surjection $\mathbb{Z}_+ \to \mathbb{Z}_+ \times \mathbb{Z}_+$. (The simultaneous choice of the f_n uses the axiom of choice.) ■

Theorem 1.16 (7.6)

A finite product of countable sets is countable.

Proof. $A \times B$ is the image of $\mathbb{Z}_+ \times \mathbb{Z}_+$ under $(n, m) \mapsto (f(n), g(m))$; induct on the number of factors. ■

Theorem 1.17 (7.7)

$\{0, 1\}^\omega$ is uncountable.

Proof. Given $g : \mathbb{Z}_+ \to \{0, 1\}^\omega$, write $g(n) = (x_{n1}, x_{n2}, \dots)$ and let $y_n = 1 - x_{nn}$. Then $y = (y_n)$ differs from $g(n)$ in the n-th coordinate, so g is not surjective. ■

Theorem 1.18 (7.8)

For any set A there is no injection $\mathcal{P}(A) \to A$ and no surjection $A \to \mathcal{P}(A)$.

Proof. An injection $\mathcal{P}(A) \to A$ would yield a surjection $A \to \mathcal{P}(A)$. Given $g : A \to \mathcal{P}(A)$, let $B = \{a : a \notin g(a)\}$. If $B = g(a_0)$, then $a_0 \in B \iff a_0 \notin g(a_0) = B$, a contradiction. ■

1.8 The principle of recursive definition

Motivation. Sequences are often built step by step, each term depending on the previous ones. The recursion principle justifies such definitions rigorously, and it is used every time we "choose x_1, then x_2, ...".

Fix a set C and the requirement $(*)$: $h(1) =$ the least element of C, and $h(i) =$ the least element of $C - h(\{1, \dots, i-1\})$ (with $C \subset \mathbb{Z}_+$ infinite, as in Lemma 1.12); the general case is identical.

Lemma 1.19 (8.1)

For each n there is a function $f : \{1, \dots, n\} \to C$ satisfying $(*)$ on its domain.

Proof. Induction: extend a function on $\{1, \dots, n\}$ by the value prescribed by $(*)$ at $n+1$ (the set $C - f(\{1, \dots, n\})$ is nonempty since C is infinite). ■

Lemma 1.20 (8.2)

Two functions on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ satisfying $(*)$ agree on their common domain.

Proof. If not, let i be the least point of disagreement; $(*)$ determines the value at i from the (equal) values below i. ■

Theorem 1.21 (8.3–8.4, principle of recursive definition)

Let A be a set, $a_0 \in A$, and ρ a function assigning to each function f from a nonempty section $\{1, \dots, n\}$ into A an element $\rho(f) \in A$. Then there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_0$ and $h(i) = \rho(h|_{\{1,\dots,i-1\}})$ for $i > 1$.

Proof. As in Lemmas 1.19 and 1.20, for each n there is a unique function on $\{1, \dots, n\}$ satisfying the formula, and these are compatible; their union is a function h on $\mathbb{Z}_+$ with the property, unique by the same argument. ■

1.9 Infinite sets and the axiom of choice

Motivation. Some infinite constructions require infinitely many arbitrary choices with no rule to make them. The axiom of choice makes this legitimate; we isolate where it is needed, for instance to show that every infinite set contains a countably infinite subset.

Theorem 1.22 (9.1)

For a set A the following are equivalent:
(1) there is an injection $\mathbb{Z}_+ \to A$;
(2) there is a bijection of A with a proper subset of itself;
(3) A is infinite.

Proof. (1)⇒(2): if $f : \mathbb{Z}_+ \to A$ is injective, define $g(f(n)) = f(n+1)$ and $g(x) = x$ off $f(\mathbb{Z}_+)$; g is a bijection onto $A - \{f(1)\}$.

(2)⇒(3) is Corollary 1.7. (3)⇒(1): let c be a choice function on the nonempty subsets of A (Lemma 1.23) and define recursively $f(1) = c(A)$, $f(n) = c(A - f(\{1, \dots, n-1\}))$; the sets are nonempty because A is infinite, and f is injective. ■

Definition (Axiom of choice)

Given a collection $\mathcal{A}$ of disjoint nonempty sets, there is a set C consisting of exactly one element from each member of $\mathcal{A}$.

Lemma 1.23 (9.2, choice function)

Given a collection $\mathcal{B}$ of nonempty sets (not necessarily disjoint), there is a function $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$ for each B.

Proof. The sets $B' = \{B\} \times B$, $B \in \mathcal{B}$, are disjoint and nonempty. A set C meeting each B' in one point is a set of pairs (B, x) with exactly one pair for each B; it is the desired function. ■

1.10 Well-ordered sets

Motivation. Well-ordered sets allow induction beyond the integers. The minimal uncountable well-ordered set S_Ω built here is the source of many counterexamples later in the course.

An ordered set is **well-ordered** if every nonempty subset has a smallest element. Subsets of well-ordered sets are well-ordered, and so are dictionary-ordered finite products of well-ordered sets. For α in a well-ordered set X, the **section** is $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.24 (10.1)

Every nonempty finite ordered set has the order type of a section $\{1, \dots, n\}$; hence it is well-ordered.

Proof. Induction on the cardinality: remove the largest element (which exists in a finite ordered set, by induction) and append it at $n+1$. ■

Theorem 1.25 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (Equivalent to the axiom of choice.)

We accept this; a proof is a transfinite version of Theorem 1.22 using a choice function. In particular there exist uncountable well-ordered sets.

Lemma 1.26 (10.2)

There is a well-ordered set A with a largest element Ω such that the section S_Ω is uncountable but every other section of A is countable.

Proof. Let B be an uncountable well-ordered set and $C = \{1, 2\} \times B$ in dictionary order; some section of C is uncountable (e.g. that of $2 \times b$). Let Ω be the least element of C whose section is uncountable, and $A = S_\Omega \cup \{\Omega\}$. ■

S_Ω is called the **minimal uncountable well-ordered set**, and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$.

Theorem 1.27 (10.3)

Every countable subset A of S_Ω has an upper bound in S_Ω.

Proof. $\bigcup_{a \in A} S_a$ is a countable union of countable sets, hence countable, so it is not all of S_Ω; any x outside it is an upper bound for A. ■

1.11 The maximum principle

Motivation. Zorn's lemma is the tool for proving that maximal objects exist (maximal ideals, bases, maximal filters); it is equivalent to the axiom of choice and will be used in the proof of the Tychonoff theorem.

A **strict partial order** $\prec$ is nonreflexive and transitive; a subset on which it is a simple order is a **chain**.

Theorem 1.28 (11.1, Hausdorff maximum principle)

In a set with a strict partial order there is a maximal chain (a chain not properly contained in any other chain).

Proof. Well-order the set A by $<$ and define by transfinite recursion $h(\alpha) = \mathit{true}$ iff $\{\beta < \alpha : h(\beta) = \mathit{true}\} \cup \{\alpha\}$ is a chain for $\prec$. The set B of α with $h(\alpha)$ true is a chain (any two of its elements lie in the chain tested at the later one). It is maximal: if $B \cup \{\alpha\}$ were a chain for some $\alpha \notin B$, then $\{\beta < \alpha : \beta \in B\} \cup \{\alpha\}$ would be a chain, forcing $\alpha \in B$. ■

Theorem 1.29 (Zorn's lemma)

If A is strictly partially ordered and every chain has an upper bound in A, then A has a maximal element.

Proof. Take a maximal chain B and an upper bound c of B. If $c \prec d$ for some d, then $B \cup \{d\}$ is a chain properly containing B (note $d \notin B$ since $d \succ c \succeq$ every element of B), contradicting maximality. ■

Example 1.1

Every vector space has a basis: the linearly independent sets, ordered by proper inclusion, satisfy the hypothesis of Zorn's lemma (a union of a chain is independent), and a maximal independent set spans. Zorn's lemma, the well-ordering theorem, the Hausdorff maximum principle and the axiom of choice are all equivalent.